

EXTENSION 1

Stage 6

Functions (Ext1), F1 Further Work With Functions (Y11)

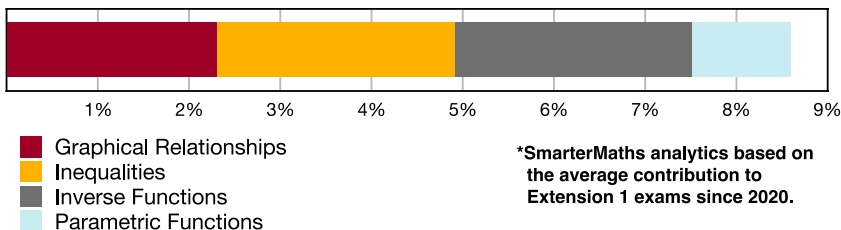
Parametric Functions (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 46.5 minutes (based on allocation of 1.5 minutes per mark)



F1 Further Work With Functions



HISTORICAL CONTRIBUTION

- F1 Further Work With Functions* has contributed a healthy 8.6% per new syllabus Ext1 exam since it was introduced in 2020.
- This topic has been split into four categories for analysis purposes which are: 1-*Graphical Relationships* (2.3%), 2-*Inequalities* (2.6%), 3-*Inverse Functions* (2.6%) and 4-*Parametric Functions* (1.1%).
- This analysis looks at *Parametric Functions*.

HSC ANALYSIS - What to expect and common pitfalls

- Parametric Equations* (1.1%) looks at linear, quadratic and circle equations in the context of parametric representation.
- It has been examined by a low band 2-mark question in 2023 and in the multiple choice section of the 2021 and 2022 exams.
- 2021 Ext1 8 MC involved the parametric equation of a semi-circle caused problems for most students, producing a mean mark of just 31%.
- The most common question type requires students to convert equations from parametric form into Cartesian form (see 2023 Ext1 11a) and then possibly sketch the resulting graph.
- A less common question type will require the opposite conversion from Cartesian to parametric form. We recommend students pay attention to Ext1 F1 SM-Bank 8 which looks at this conversion for a circle equation.

Questions

1. Functions, EXT1 F1 2019 SPEC2-N 2 MC

The curve given by $x = 3\sec(t) + 1$ and $y = 2\tan(t) - 1$ can be expressed in cartesian form as

- A. $\frac{(y+1)^2}{4} - \frac{(x-1)^2}{9} = 1$
- B. $\frac{(x+1)^2}{3} - \frac{(y-1)^2}{2} = 1$
- C. $\frac{(x-1)^2}{3} + \frac{(y+1)^2}{2} = 1$
- D. $\frac{(x-1)^2}{9} - \frac{(y+1)^2}{4} = 1$

2. Functions, EXT1 F1 EQ-Bank 4 MC

Determine the Cartesian equation of the circle given by the parametric equations

$$x = -3 + 4\cos\theta$$

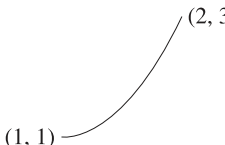
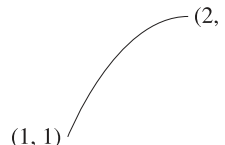
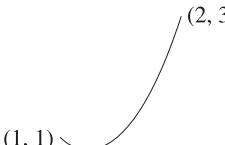
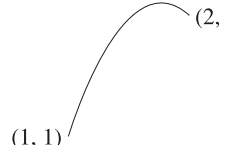
$$y = 1 + 4\sin\theta$$

- A. $(x+3)^2 + (y-1)^2 = 4$
- B. $(x-3)^2 + (y+1)^2 = 4$
- C. $(x+3)^2 + (y-1)^2 = 16$
- D. $(x-3)^2 + (y+1)^2 = 16$

3. Functions, EXT1 F1 2022 HSC 5 MC

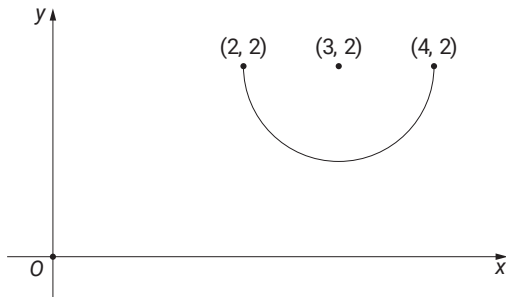
A curve is defined in parametric form by $x = 2 + t$ and $y = 3 - 2t^2$ for $-1 \leq t \leq 0$.

Which diagram best represents this curve?

- A. 
- B. 
- C. 
- D. 

4. Functions, EXT1 F1 2021 HSC 8 MC

The diagram shows a semicircle.



Which pair of parametric equations represents the semicircle shown?

- A. $\begin{cases} x = 3 + \sin t \\ y = 2 + \cos t \end{cases}$ for $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$
- B. $\begin{cases} x = 3 + \cos t \\ y = 2 + \sin t \end{cases}$ for $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$
- C. $\begin{cases} x = 3 - \sin t \\ y = 2 - \cos t \end{cases}$ for $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$
- D. $\begin{cases} x = 3 - \cos t \\ y = 2 - \sin t \end{cases}$ for $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$

5. Functions, EXT1 F1 2023 HSC 11a

The parametric equations of a line are given below.

$$\begin{aligned} x &= 1 + 3t \\ y &= 4t \end{aligned}$$

Find the Cartesian equation of this line in the form $y = mx + c$. (2 marks)

6. Functions, EXT1 F1 SM-Bank 4

Sketch the curve described by these two parametric equations

$$\begin{aligned} x &= 3\cos t + 2 \\ y &= 3\sin t - 3 \text{ for } 0 \leq t < 2\pi. \end{aligned} \quad (3 \text{ marks})$$

7. Functions, EXT1 F1 SM-Bank 5

Find the Cartesian equation for the curve with the parametric equations

$$\begin{aligned} x &= 3t \\ y &= 2t^2. \end{aligned} \quad (1 \text{ mark})$$

8. Functions, EXT1 F1 EQ-Bank 6

The parametric equations of a graph are

$$\begin{aligned} x &= t^2 \\ y &= \frac{1}{t} \text{ for } t > 0 \end{aligned}$$

- i. Find the Cartesian equation for the graph. (1 mark)
- ii. Sketch the graph. (1 mark)

9. Functions, EXT1 F1 EQ-Bank 7

The parametric equations of a graph are

$$\begin{aligned} x &= 1 - \frac{1}{t} \\ y &= 1 + \frac{1}{t} \end{aligned}$$

Sketch the graph. (2 marks)

10. Functions, EXT1 F1 EQ-Bank 10

An equation can be expressed in the parametric form

$$\begin{aligned} x &= 2\cos\theta - 1 \\ y &= 2 + 2\sin\theta \end{aligned}$$

- i. Express the equation in Cartesian form. (2 marks)
- ii. Sketch the graph. (1 mark)

11. Functions, EXT1 F1 EQ-Bank 11

- i. Find the function described by the following parametric equations

$$\begin{aligned} x &= 3t^2 \\ y &= 9t, \quad t > 0 \end{aligned} \quad (1 \text{ mark})$$

- ii. Sketch the function. (1 mark)

12. Functions, EXT1 F1 SM-Bank 8

A circle has the equation $x^2 - 10x + y^2 + 6y + 25 = 0$

- i. Express the circle in parametric form. (2 marks)
- ii. Sketch the circle. (1 mark)

13. Functions, EXT1 F1 2003 HSC 1d

A curve has parametric equations $x = \frac{t}{2}, y = 3t^2$.

Find the Cartesian equation for this curve. **(2 marks)**

14. Functions, EXT1 F1 SM-Bank 9

i. Sketch the graph of the function described by the parametric equations

$$x = 4t - 7$$

$$y = 2t^2 + t \quad \textbf{(2 marks)}$$

ii. State the domain and range of the function. **(1 mark)**

15. Functions, EXT1 F1 2021 SPEC2 7

A function is defined parametrically by

$$x(t) = 5\cos(2t) + 1, \quad y(t) = 5\sin(2t) - 1$$

If $A(6, -1)$ and $B(1, 4)$ are two points that lie on the graph of the function, then find the shortest distance along the graph from A to B . **(2 marks)**

16. Functions, EXT1 F1 2002 HSC 1e

The variable point $(3t, 2t^2)$ lies on a parabola.

Find the Cartesian equation for this parabola. **(2 marks)**

Worked Solutions

1. Functions, EXT1 F1 2019 SPEC2-N 2 MC

$$\sec^2\theta = \tan^2\theta + 1$$

$$x = 3\sec(t) + 1 \Rightarrow \sec(t) = \frac{x-1}{3}$$

$$y = 2\tan(t) - 1 \Rightarrow \tan(t) = \frac{y+1}{2}$$

$$\therefore \left(\frac{x-1}{3}\right)^2 - \left(\frac{y+1}{2}\right)^2 = 1$$

$$\frac{(x-1)^2}{9} - \frac{(y+1)^2}{4} = 1$$

$$\Rightarrow D$$

2. Functions, EXT1 F1 EQ-Bank 4 MC

$$x = -3 + 4\cos\theta \Rightarrow \cos\theta = \frac{x+3}{4}$$

$$y = 1 + 4\sin\theta \Rightarrow \sin\theta = \frac{y-1}{4}$$

Using $\sin^2\theta + \cos^2\theta = 1$:

$$\left(\frac{x+3}{4}\right)^2 + \left(\frac{y-1}{4}\right)^2 = 1$$

$$(x+3)^2 + (y-1)^2 = 16$$

$$\Rightarrow C$$

3. Functions, EXT1 F1 2022 HSC 5 MC

$$x = 2 + t \Rightarrow t = x - 2$$

Substitute into $y = 3 - 2t^2$

$$y = 3 - 2(x-2)^2$$

Concave down parabola with maximum at $(2, 3)$

$$\Rightarrow B$$

4. Functions, EXT1 F1 2021 HSC 8 MC

By elimination:

When $t = \frac{\pi}{2}$,

$A(4, 2), B(3, 3), C(2, 2), D(3, 1)$

→ Eliminate B

When $t = -\frac{\pi}{2}$,

$A(2, 2), C(4, 2), D(3, 3)$

→ Eliminate D

When $t = 0$,

$A(3, 3), C(3, 1)$

→ Eliminate A

$\Rightarrow C$

5. Functions, EXT1 F1 2023 HSC 11a

$x = 1 + 3t \Rightarrow t = \frac{x - 1}{3}$

$y = 4t$

$y = 4\left(\frac{x - 1}{3}\right)$

$y = \frac{4}{3}x - \frac{4}{3}$

♦♦ Mean mark 31%.

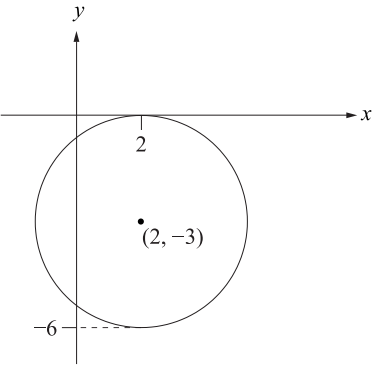
6. Functions, EXT1 F1 SM-Bank 4

$(x-2) = 3\cos t$

$(y + 3) = 3\sin t$

$$\begin{aligned}(x-2)^2 + (y + 3)^2 &= (3\cos t)^2 + (3\sin t)^2 \\ &= 9\cos^2 t + 9\sin^2 t \\ &= 9(\cos^2 t + \sin^2 t) \\ &= 9\end{aligned}$$

Sketch: $(x-2)^2 + (y + 3)^2 = 3^2$



7. Functions, EXT1 F1 SM-Bank 5

$x = 3t \dots (1)$

$y = 2t^2 \dots (2)$

Substitute $t = \frac{x}{3}$ from (1) into (2):

$y = 2\left(\frac{x}{3}\right)^2$

$y = \frac{2}{9}x^2$

8. Functions, EXT1 F1 EQ-Bank 6

i. $x = t^2 \dots (1)$
 $y = \frac{1}{t} \dots (2)$

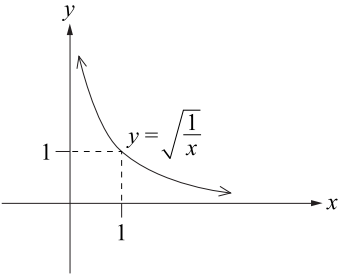
Substitute $t = \frac{1}{y}$ from (1) into (1)

$$x = \left(\frac{1}{y}\right)^2$$

$$y^2 = \frac{1}{x}$$

$$y = \sqrt{\frac{1}{x}} \quad (y > 0 \text{ as } t > 0)$$

ii.



9. Functions, EXT1 F1 EQ-Bank 7

$$x = 1 - \frac{1}{t} \dots (1)$$

$$tx = t - 1$$

$$t(1 - x) = 1$$

$$t = \frac{1}{1 - x} \dots (1')$$

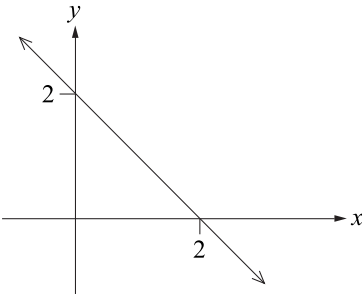
$$y = 1 + \frac{1}{t} \dots (2)$$

Substitute $t = \frac{1}{1 - x}$ from (1') into (2):

$$y = 1 + \frac{1}{\frac{1}{1 - x}}$$

$$y = 1 + 1 - x$$

$$y = 2 - x$$



10. Functions, EXT1 F1 EQ-Bank 10

i. $2\cos\theta = x + 1$

$$\cos\theta = \frac{x + 1}{2}$$

$$2\sin\theta = y - 2$$

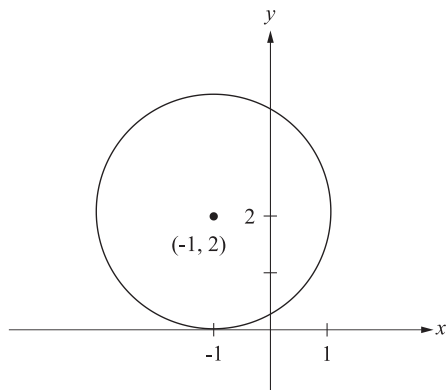
$$\sin\theta = \frac{y - 2}{2}$$

Using $\cos^2\theta + \sin^2 = 1$:

$$\left(\frac{x + 1}{2}\right)^2 + \left(\frac{y - 2}{2}\right)^2 = 1$$

$$(x + 1)^2 + (y - 2)^2 = 4$$

ii. Sketch circle with centre $(-1, 2)$, radius = 2



11. Functions, EXT1 F1 EQ-Bank 11

i. $y = 9t \Rightarrow t = \frac{y}{9}$

$$x = 3t^2$$

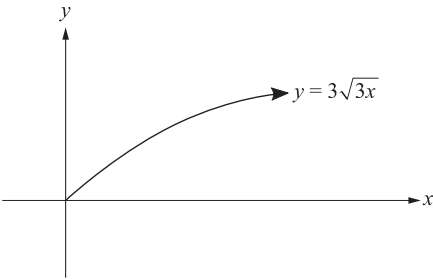
$$x = 3 \cdot \left(\frac{y}{9}\right)^2$$

$$x = \frac{y^2}{27}$$

$$y^2 = 27x$$

$$y = 3\sqrt{3x}, \quad (t > 0)$$

ii.



12. Functions, EXT1 F1 SM-Bank 8

i. $x^2 - 10x + y^2 + 6y + 25 = 0$

$$(x-5)^2 + (y+3)^2 - 9 = 0$$

$$(x-5)^2 + (y+3)^2 = 9$$

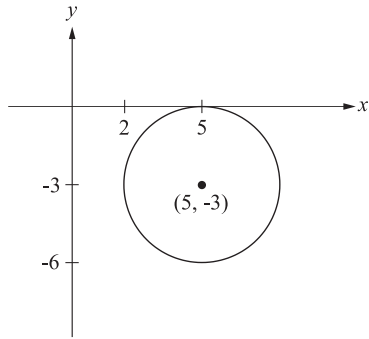
$\Rightarrow$ Circle centre $(5, -3)$, radius 3

$\therefore$ Parametric form is:

$$x = 5 + 3\cos\theta$$

$$y = -3 + 3\sin\theta$$

ii.



13. Functions, EXT1 F1 2003 HSC 1d

$$x = \frac{t}{2} \Rightarrow t = 2x$$

Substitute $t = 2x$ into $y = 3t^2$

$$y = 3(2x)^2$$

$$y = 12x^2$$

14. Functions, EXT1 F1 SM-Bank 9

i. $x = 4t - 7 \Rightarrow t = \frac{x+7}{4}$

$$y = 2t^2 + t$$

$$y = 2\left(\frac{x+7}{4}\right)^2 + \left(\frac{x+7}{4}\right)$$

$$16y = 2(x+7)^2 + 4(x+7)$$

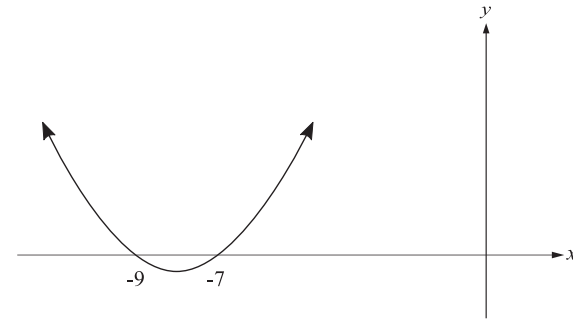
$$16y = 2x^2 + 28x + 98 + 4x + 28$$

$$16y = 2x^2 + 32x + 126$$

$$8y = x^2 + 16x + 63$$

$$y = \frac{1}{8}(x+7)(x+9)$$

$\Rightarrow$ Equation is a concave up quadratic with zeros at $x = -9$ and $x = -7$.



ii. Axis at $x = -8$

$$\begin{aligned} \therefore y_{\min} &= \frac{1}{8}(-1)(1) \\ &= -\frac{1}{8} \end{aligned}$$

Domain: all x

Range: $-\frac{1}{8} \leq y < \infty$

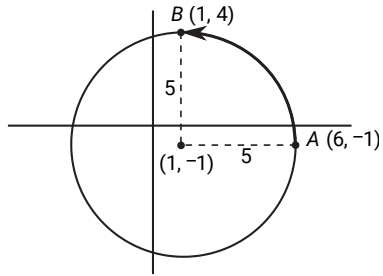
15. Functions, EXT1 F1 2021 SPEC2 7

$$x = 5\cos(2t) + 1 \Rightarrow \cos(2t) = \frac{x-1}{5}$$

$$y = 5\sin(2t) - 1 \Rightarrow \sin(2t) = \frac{y+1}{5}$$

$$(\cos(2t))^2 + (\sin(2t))^2 = 1$$

$$(x-1)^2 + (y+1)^2 = 25$$



$$\begin{aligned} \text{Distance} &= \frac{1}{4} \times 2 \times \pi \times r \\ &= \frac{5\pi}{2} \end{aligned}$$

16. Functions, EXT1 F1 2002 HSC 1e

$$x = 3t \Rightarrow t = \frac{x}{3}$$

Substitute $t = \frac{x}{3}$ into $y = 2t^2$

$$y = 2 \times \left(\frac{x}{3}\right)^2$$

$$y = \frac{2x^2}{9}$$